Figure 8.19 Comparing memories for the same information given two different memory types

Memory and knowledge can significantly assist an agent with various application types. Indeed, a single memory/knowledge store could feed one or multiple agents, allowing for further specialized interpretations of both types of stores. We'll finish out the chapter by discussing memory/knowledge compression next.

8.7 Understanding memory and knowledge compression

Much like our own memory, memory stores can become cluttered with redundant information and numerous unrelated details over time. Internally, our minds deal with memory clutter by compressing or summarizing memories. Our minds remember more significant details over less important ones, and memories accessed more frequently.

We can apply similar principles of memory compression to agent memory and other retrieval systems to extract significant details. The principle of compression is similar to semantic augmentation but adds another layer to the preclusters groups of related memories that can collectively be summarized.

Figure 8.20 shows the process of memory/knowledge compression. Memories or knowledge are first clustered using an algorithm such as k-means. Then, the groups of memories are passed through a compression function, which summarizes and collects the items into more succinct representations.